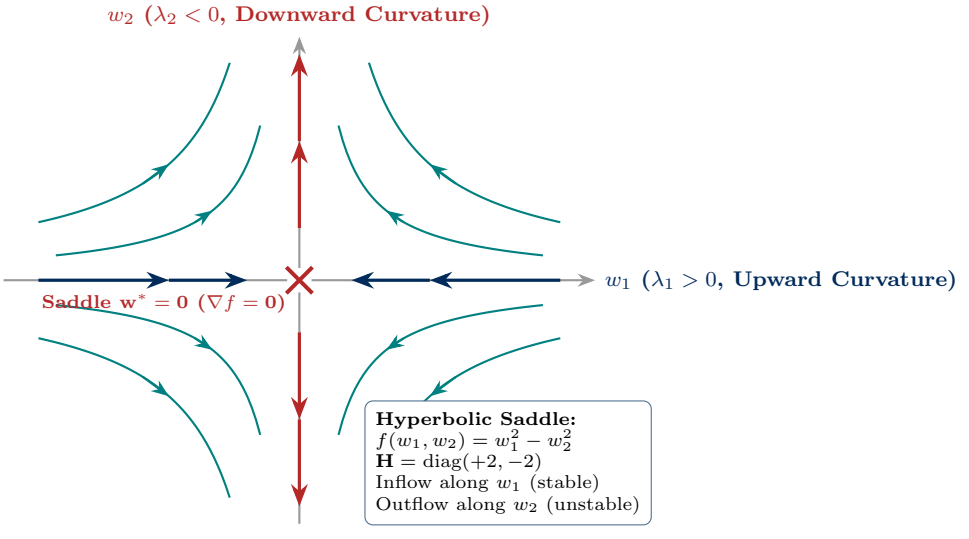

w_2 ($\lambda_2 < 0$, Downward Curvature)
w_1 ($\lambda_1 > 0$, Upward Curvature)
Saddle $\mathbf{w}^* = \mathbf{0}$ ($\nabla f = \mathbf{0}$)
Hyperbolic Saddle:
$f(w_1, w_2) = w_1^2 - w_2^2$
$\mathbf{H} = \mathrm{diag}(+2, -2)$
Inflow along w_1 (stable)
Outflow along w_2 (unstable)